AN ELEMENTARY PROOF OF ERLBACHER'S COUNTEREXAMPLE TO TXGRAFFITI CONJECTURE 9

ABSTRACT. Erlbacher identified and computationally verified a connected cubic triangle-free graph on 14 vertices that refutes Conjecture 9 of Davila, generated by TxGraffiti. We give a self-contained elementary proof that its domination number is 4 and its zero forcing number is 7. Hence $Z(G) = \gamma(G) + 3$, contradicting the proposed bound $Z(G) \leq \gamma(G) + 2$ for connected cubic diamond-free graphs. The lower bound on $Z(G)$ follows from an explicit family of forts.

1. The counterexample

All graphs are finite, simple, and undirected. A set $D \subseteq V(G)$ is *dominating* if $N[D] = V(G)$, and the minimum size of such a set is the *domination number* $\gamma(G)$. Starting from a set of blue vertices, the zero forcing rule allows a blue vertex with exactly one white neighbor to force that neighbor blue. The minimum size of an initial set that can color all vertices blue is the *zero forcing number* $Z(G)$; see [1].

Davila posed the following as Conjecture 9 of [2], generated by TxGraffiti:

$$Z(G) \leq \gamma(G) + 2 \tag{1}$$

for every connected cubic diamond-free graph G. Erlbacher [3] exhibited the graph below and verified its parameters by exhaustive computation. The same artifact gives larger computational examples in a chain family. The contribution here is a noncomputational proof for the 14-vertex member.

Let

$$A = \{a_1, a_2, a_3\}, \quad B = \{b_1, b_2, b_3\},$$
$$C = \{c_1, c_2, c_3\}, \quad D = \{d_1, d_2, d_3\}.$$

Define G on $A \cup B \cup C \cup D \cup \{x, y\}$ by

$$\begin{aligned} E(G) = &\{a_i b_j : 1 \leq i, j \leq 3,\ (i,j) \neq (3,1)\} \\ &\cup \{c_i d_j : 1 \leq i, j \leq 3,\ (i,j) \neq (3,1)\} \\ &\cup \{a_3 x, x b_1, c_3 y, y d_1, xy\}. \end{aligned}$$

Equivalently, take two copies of $K_{3,3}$, subdivide one edge in each, and join the two subdivision vertices.

2020 *Mathematics Subject Classification.* 05C50, 05C69.

Key words and phrases. zero forcing number, domination number, cubic graph, fort, counterexample.

Theorem 1. *The graph G is connected, cubic, and triangle-free, and*

$$\gamma(G) = 4 \qquad \text{and} \qquad Z(G) = 7.$$

Consequently, G is diamond-free and refutes (1).

A nonempty set $F \subseteq V(G)$ is a *fort* if every vertex outside F has either zero or at least two neighbors in F.

Lemma 2. *Every zero forcing set intersects every fort.*

Proof. If a zero forcing set S were disjoint from a fort F, then immediately before the first force into F, the forcing vertex would lie outside F and have exactly one neighbor in F, contrary to the definition of a fort. □

Proof of Theorem 1. The construction makes G connected and cubic. Each subdivided copy of $K_{3,3}$ is triangle-free, and the edge xy is a bridge, so G is triangle-free. Since every diamond contains a triangle, G is diamond-free.

The set $\{a_3, b_1, c_3, d_1\}$ dominates G, so $\gamma(G) \leq 4$. Since G has 14 vertices and maximum degree 3, a set of k vertices dominates at most $4k$ vertices. Hence $\gamma(G) \geq \lceil 14/4 \rceil = 4$, and therefore $\gamma(G) = 4$.

For the zero forcing upper bound, let

$$S_0 = \{y, a_1, a_2, b_2, c_1, c_2, d_2\}.$$

The sequence

$$b_2 \to a_3, \quad d_2 \to c_3, \quad c_3 \to d_3, \quad c_1 \to d_1, \quad y \to x, \quad a_3 \to b_3, \quad a_1 \to b_1$$

is valid, so $Z(G) \leq 7$.

It remains to prove $Z(G) \geq 7$. Put $U = A \cup B$ and $W = C \cup D$. The following six subsets of U are forts:

$$\{a_1, a_2\}, \qquad \{b_2, b_3\},$$

and

$$\{a_i, a_3, b_1, b_j\}, \qquad i \in \{1, 2\}, \quad j \in \{2, 3\}.$$

For either two-element set, each outside vertex adjacent to it has exactly two neighbors in it. For a four-element set, the only outside vertices with neighbors in the set are x, the other vertex of $\{a_1, a_2\}$, and the other vertex of $\{b_2, b_3\}$; each has exactly two such neighbors.

No two vertices meet all six forts. Indeed, any two-vertex set meeting the first two forts must be $\{a_i, b_j\}$ with $i \in \{1, 2\}$ and $j \in \{2, 3\}$, but it misses the four-element fort obtained from the other choices of i and j. Lemma 2 therefore gives

$$|S \cap U| \geq 3$$

for every zero forcing set S. By symmetry, $|S \cap W| \geq 3$.

Suppose that S is a zero forcing set of size six. Then

$$|S \cap U| = |S \cap W| = 3 \qquad \text{and} \qquad S \cap \{x, y\} = \varnothing.$$

Define

$$\mathcal{P} = \{\{a_1, a_3\}, \{a_2, a_3\}, \{b_1, b_2\}, \{b_1, b_3\}\},$$
$$\mathcal{Q} = \{\{c_1, c_3\}, \{c_2, c_3\}, \{d_1, d_2\}, \{d_1, d_3\}\}.$$

For every $p \in \mathcal{P}$ and $q \in \mathcal{Q}$, the set

$$\{x, y\} \cup p \cup q$$

is a fort. If $p = \{a_i, a_3\}$, the other vertex of $\{a_1, a_2\}$ has zero neighbors in $p \cup \{x\}$, while b_1, b_2, b_3 each have two. If $p = \{b_1, b_j\}$, the other vertex of $\{b_2, b_3\}$ has zero, while a_1, a_2, a_3 each have two. The verification for q is symmetric.

Since S misses x and y, Lemma 2 implies that S meets p or q for every $p \in \mathcal{P}$ and $q \in \mathcal{Q}$. Hence S meets every member of $\mathcal{P}$, or it meets every member of $\mathcal{Q}$. In the first case, together with the two two-element forts above, $S \cap U$ meets all six pairs

$$\{a_1, a_2\},\ \{a_1, a_3\},\ \{a_2, a_3\}, \qquad \{b_1, b_2\},\ \{b_1, b_3\},\ \{b_2, b_3\}.$$

These are the edges of two vertex-disjoint copies of K_3, so at least two vertices from each copy are required. Thus $|S \cap U| \geq 4$, a contradiction. The second case similarly gives $|S \cap W| \geq 4$. Therefore no six-vertex set is zero forcing, so $Z(G) \geq 7$. Together with the forcing sequence, this proves $Z(G) = 7$. □

The edge xy is a bridge, but bridgelessness is not a hypothesis of Conjecture 9. Moreover, x and y are centers of induced claws, so the example does not affect Davila's theorem for connected cubic claw-free graphs.

References

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